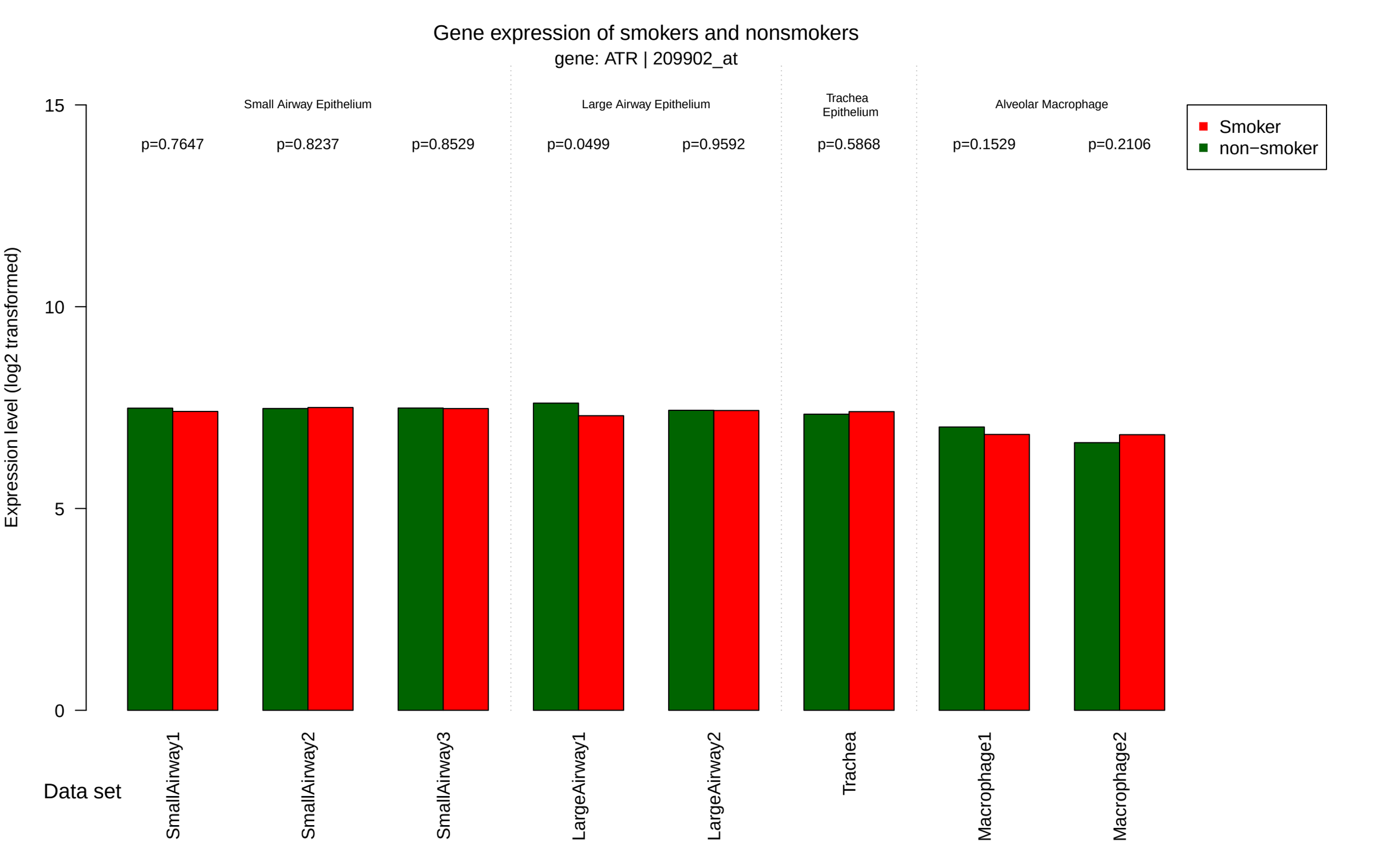
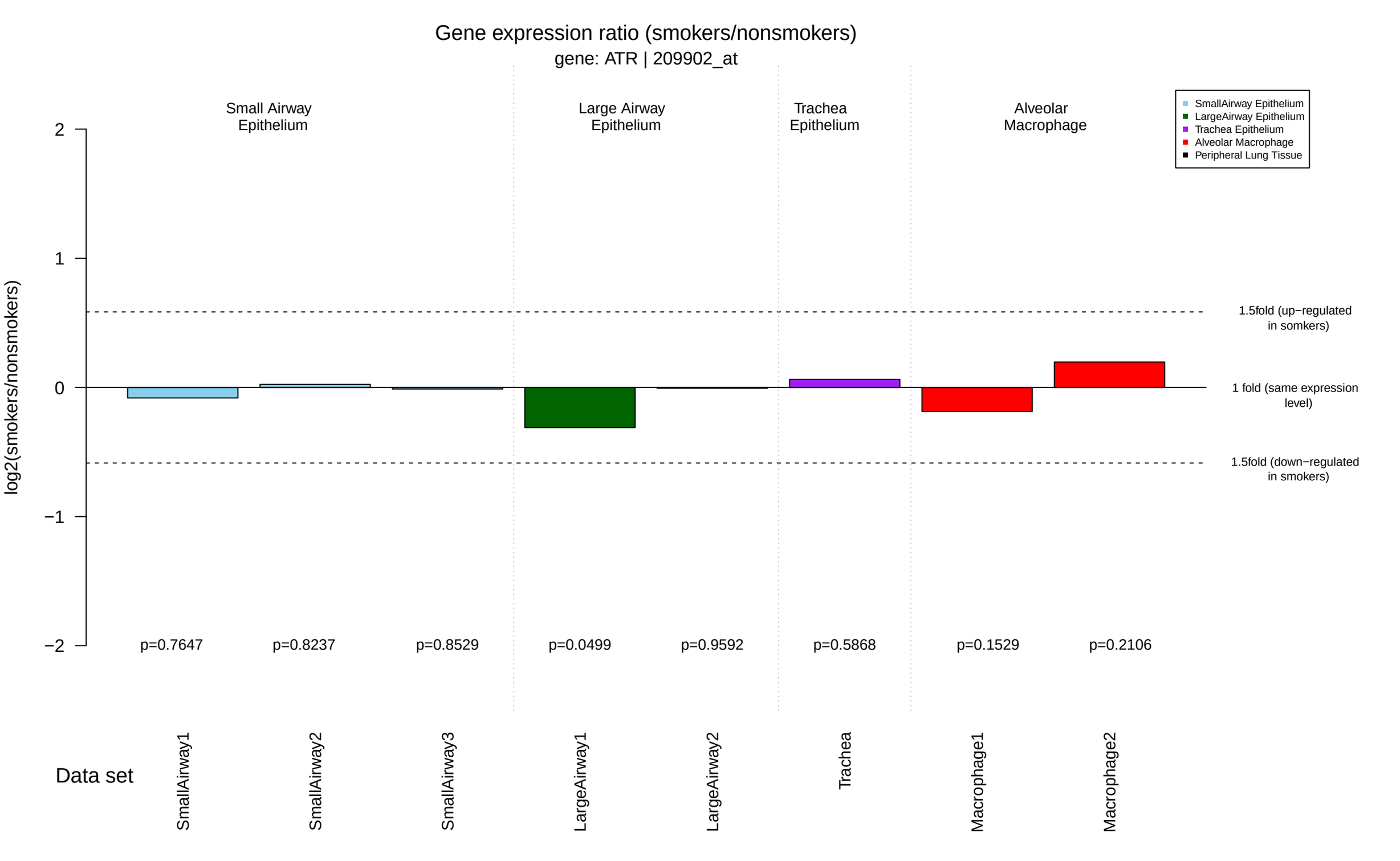
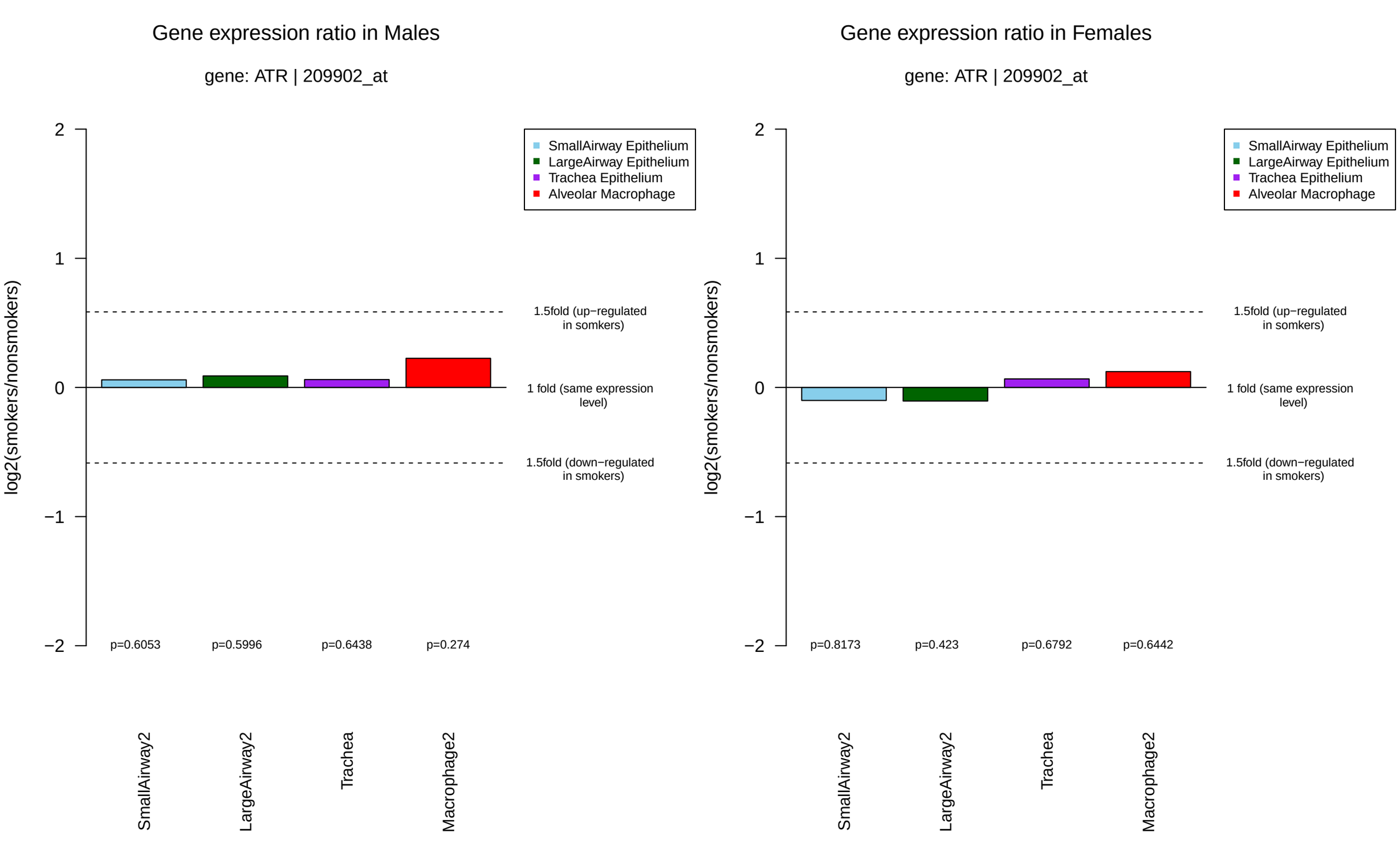
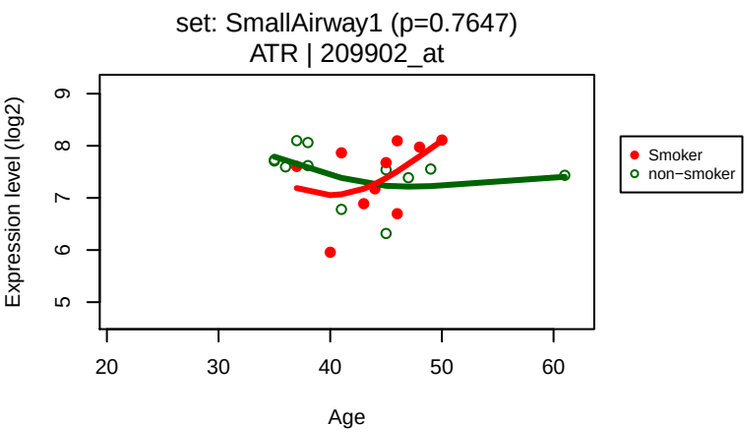
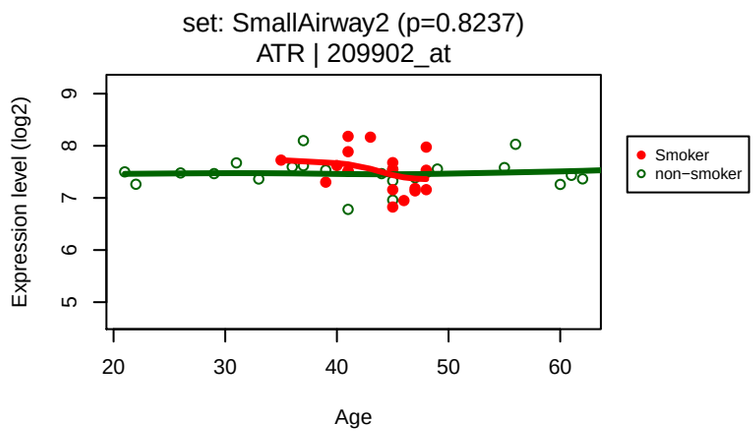
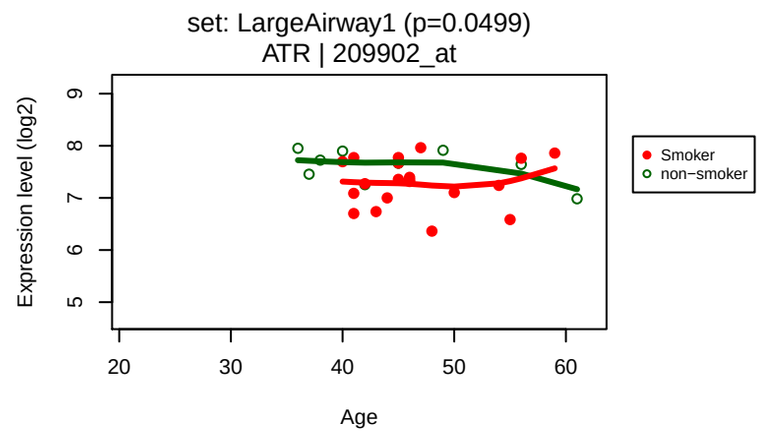

Gene expression of smokers and nonsmokers
gene: ATR | 209902_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7647
p=0.8237
p=0.8529
p=0.0499
p=0.9592
p=0.5868
p=0.1529
p=0.2106
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

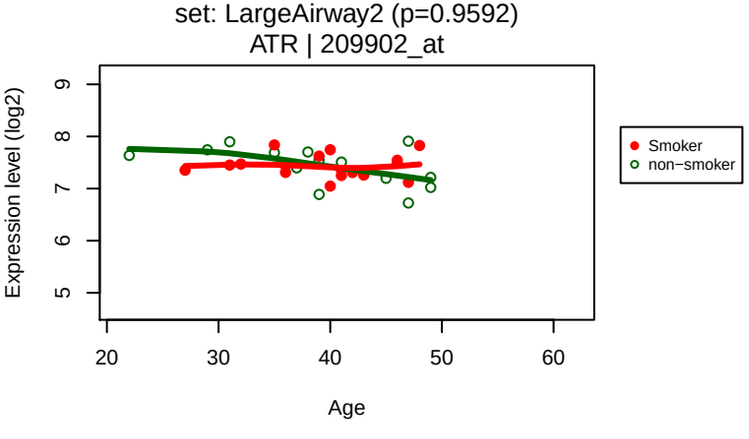

Gene expression ratio (smokers/nonsmokers)
gene: ATR | 209902_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7647
p=0.8237
p=0.8529
p=0.0499
p=0.9592
p=0.5868
p=0.1529
p=0.2106
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

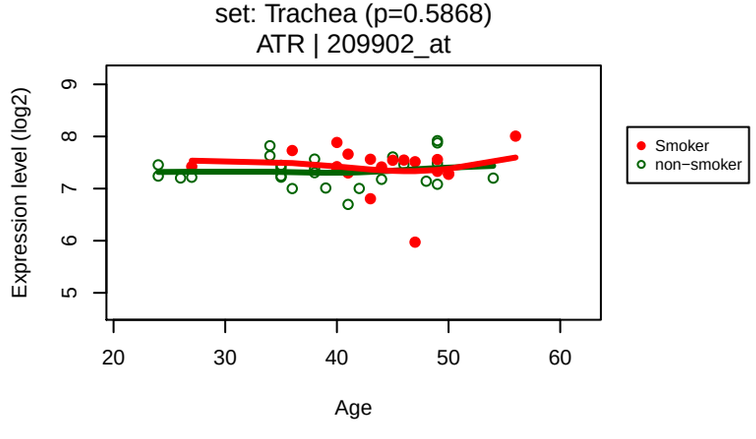

Gene expression ratio in Males
gene: ATR | 209902_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6053
p=0.5996
p=0.6438
p=0.274
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ATR | 209902_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8173
p=0.423
p=0.6792
p=0.6442
SmallAirway2
LargeAirway2
Trachea
Macrophage2

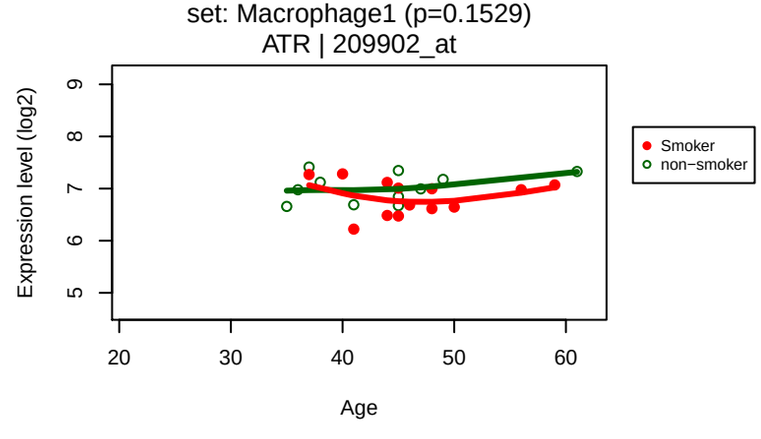

set: SmallAirway1 (p=0.7647)
ATR | 209902_at
Expression level (log2)
Age
Smoker
non−smoker

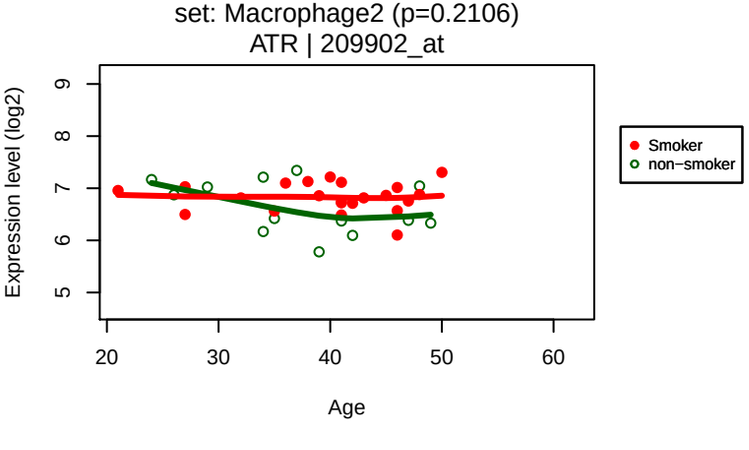

set: SmallAirway2 (p=0.8237)
ATR | 209902_at
Expression level (log2)
Age
Smoker
non−smoker

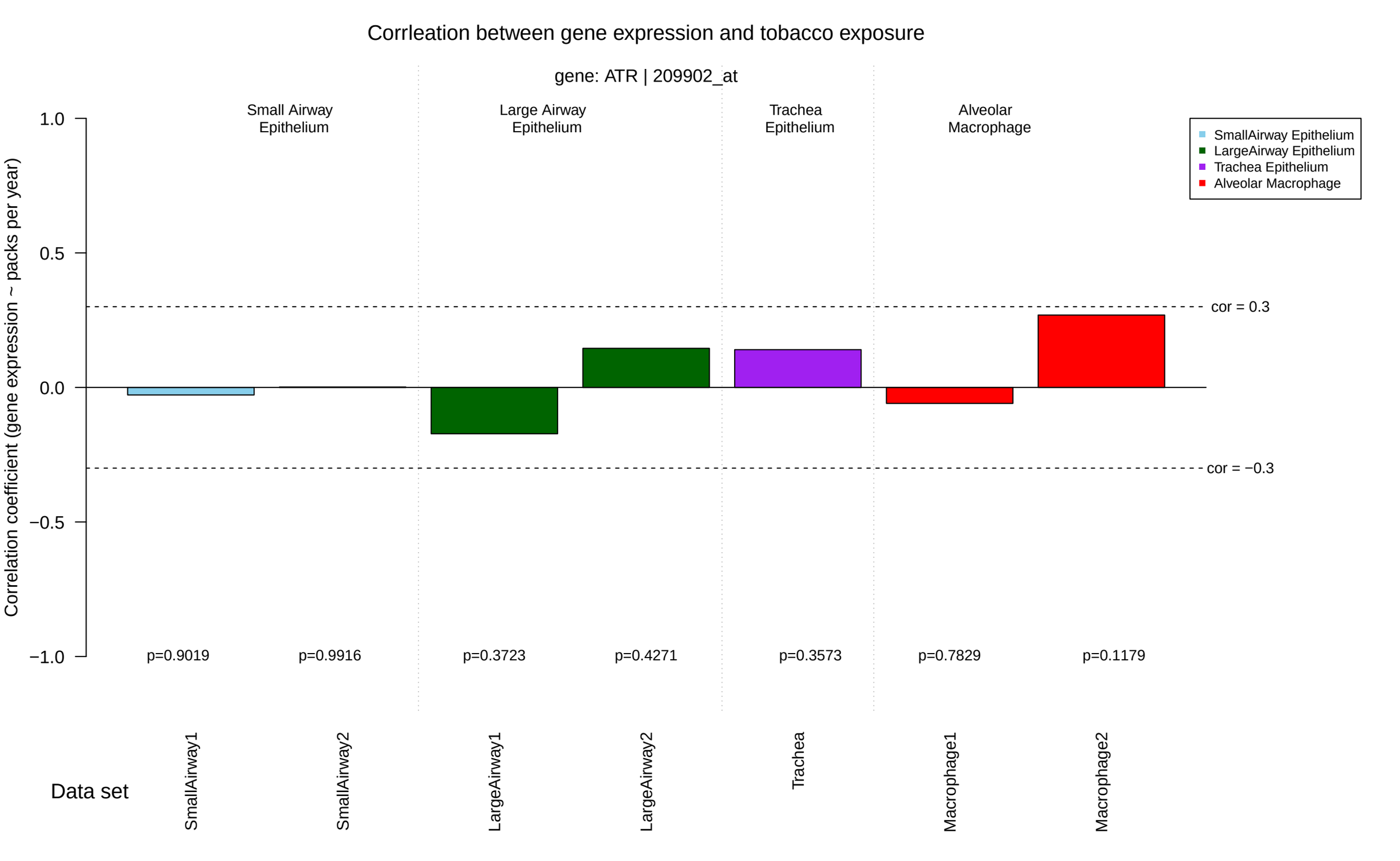

set: LargeAirway1 (p=0.0499)
ATR | 209902_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9592)
ATR | 209902_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5868)
ATR | 209902_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1529)
ATR | 209902_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2106)
ATR | 209902_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ATR | 209902_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9019
p=0.9916
p=0.3723
p=0.4271
p=0.3573
p=0.7829
p=0.1179
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2